

FIRST-PASSAGE TIMES OF INCREASING MARKOV PROCESSES AND CLASSES OF LIFE DISTRIBUTIONS

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Abstract

While modelling deterioration or ageing of devices, first-passage times of Markov processes play a significant role, especially when the devices are subject to shocks and wear during their operation. In view of this, obtaining sufficient conditions for first-passage times to belong to specific ageing families constitutes an important problem. There exists a rich literature dealing with this class of problems; see for example [11], [18], [35], [67]. We address the same problem in the context of some new ageing classes such as DMTTF (IMTTF), IMIT, and DRHR. In this connection, some issues in [11] have also been investigated. We wind up by including certain examples to highlight the practical relevance of the results.

Keywords: First-passage times; life distributions; DMTTF; IMTTF; IMIT; DRHR

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Secondary 60K20

1. Introduction

The notion of first-passage time, also called the stopping time, of a stochastic process has great significance in probability literature and as such has been studied in various contexts. For instance, an important connection between first-passage times and Green's function for a broad class of additive skip-free processes has been established in Keilson [40]. For a hyper-exponential jump diffusion process, many important probabilistic properties related to the first-passage times such as finiteness, expectation, conditional memorylessness, and conditional independence are studied in Cai [20], while the problem of obtaining the distributions of the first-passage times to flat boundaries for the said process has been investigated in Yin et al. [80]. Gitterman [29] evaluates the mean of the first-passage time, denoted as MFPT, for the anomalous diffusion, and has shown that MFPT can exhibit non-monotonic trends with the degree of departure from normal diffusion. First-passage times also have a significant role in the domain of mathematical finance (see [2], [26], [51], [52], etc.).

In reliability theory, while modelling deterioration or ageing of systems, first-passage times of Markov processes arise naturally in the context of devices subject to shocks and wear during their operation. For deterioration models, the first-passage time is simply the lifetime of a unit

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which fails whenever a critical threshold is exceeded. While obtaining the distribution of first-passage times has received a great deal of attention in the literature (see e.g. [24], [63], [74–77], etc.), it becomes of fundamental importance to analyse relevant conditions under which the distribution function of the first-passage time possesses various ageing properties pertaining to reliability and maintenance. First-passage times of Markov processes having PF_r (Pólya frequency of order r) density are investigated in Assaf et al. [5]. Brown and Chaganty [18] investigated the IFR property of first-passage times which was further studied in Durham et al. [27]. Sufficient conditions ensuring the discrete IFRA property of first-passage times were also established in Brown and Chaganty [18], whose continuous analogues were given in Shaked and Shanthikumar [72]. The NBU case has been considered by Marshall and Shaked [59, 60] and also by Shanthikumar [73]. For an increasing Markov process, the NBUE and NWUE properties were discussed in Karasu and Özekici [35]. Subsequently, Lam [54] extended their work to the framework of a general Markov renewal process and further showed that under certain conditions, a more general class of stochastic processes possess the NBUE property. NBUC and DMRL classes have been studied in Pérez-Ocón and Gámiz-Pérez [67]. Belzunce et al. [11] have considered the problem in the context of four important classes, namely IFR(2), NBU(2) and DRL_{Lt} , NBU_{Lt} , that arise from the ‘increasing concave order’ and the ‘Laplace transform order’ respectively.

The present article aims to extend the research along this direction to some new ageing classes that arise naturally depending on the monotonicity of certain reliability functions, namely the mean time to failure (MTTF) function, mean inactive time (MIT) function, and reversed hazard rate (RHR) function. In recent times, the literature in reliability has focused quite heavily on these functions, and the ageing classes derived from them are, namely DMTTF (IMTTF), IMIT, and DRHR. Various probabilistic aspects have been explored in [8], [23], [32], [33], [45], [46], [57], [64–66], [69], [70], [79], etc., while inferential issues have been dealt with in [12–16], [25], [37], [38], [43], [44], [58], [61], [62]. However, the problem of obtaining appropriate conditions in the presence of which first-passage times of a Markov process possess such ageing properties remains unexplored. The present article tries to address precisely this issue. Apart from that, the authors have also analysed certain aspects of the work of Belzunce et al. [11].

Let $Y = \{Y_t, t \geq 0\}$ be a time homogeneous Markov process with transition function P_t on a countable set $E \subset [0, +\infty]$, where Y_t represents age or the amount of deterioration or degradation of a device at time t . It is reasonable to assume (see [11] or [35]) that Y is increasing and right continuous on E . Then Y is regular and can be defined uniquely in terms of $(X, T) = \{(X_n, T_n); n \geq 0\}$, provided $\lim_{n \rightarrow \infty} T_n = +\infty$ (see Çinlar [22]), where X_n is the n th state visited by Y , and T_n is the time of its n th jump. The embedded process X is a strictly increasing Markov chain on E with an upper-triangular transition matrix Q .

We consider that the device fails whenever a critical threshold $j \in E$ is exceeded, and in this paper we are interested in the first-passage time defined by the random variable

$$S_j = \inf\{n \geq 0 \mid X_n > j\}, \quad j \in E \quad (1.1)$$

whenever the process is confined to discrete time, and similarly by

$$L_j = \inf\{t \geq 0 \mid Y_t > j\}, \quad j \in E \quad (1.2)$$

for the continuous case.

The organization of the paper is as follows. In Section 2 we recapitulate some preliminaries related to certain stochastic processes, and thereafter we present a brief introduction about the

ageing classes we will be working with. Section 3 confines itself to establishing appropriate conditions under which first-passage time of a discrete-time increasing Markov chain possesses the ageing properties DMTTF (IMTTF), IMIT, and DRHR. The analogous conditions for L_j are next developed in Section 4. Finally, in Section 5, we discuss certain implications that can be found among these sufficient conditions. These types of implications in the context of our concerned reliability classes have also been obtained in this section. We finally add some examples to demonstrate the practical relevance of the present work.

Throughout this paper, we will make it a practice to write ‘increasing’ and ‘decreasing’ instead of ‘non-decreasing’ and ‘non-increasing’, respectively.

2. Notations, definitions, and related discussion

In this section we recall some well-known definitions and lay the groundwork for what follows by presenting the necessary mathematical preliminaries.

For any matrix M , define the cumulative matrix $\bar{M}$ as

$$\bar{M}(i, j) = \sum_{l \leq j} M(i, l), \quad i, j \in E.$$

Then, if X starts at i , the survival function of S_j can be written as

$$\mathbb{P}(S_j > n \mid X_0 = i) = \mathbb{P}(X_n \leq j \mid X_0 = i) := \bar{Q}^n(i, j) \quad \text{for all } n \geq 0, \quad i, j \in E.$$

Again, for the transition matrix Q , we have the potential matrix of X defined by

$$R(i, j) = \sum_{k=0}^{\infty} Q^k(i, j).$$

Next we have the cumulative residual potential matrix, introduced in Pérez-Ocón and Gámiz-Pérez [67], defined by

$$R_n(i, j) = \sum_{k=n}^{\infty} Q^k(i, j).$$

Belzunce et al. [11] added the additional notation

$$R_{n,m}(i, j) = \sum_{k=n}^{n+m} Q^k(i, j).$$

Before proceeding further, it is pertinent to mention that in the proof of Theorem 1(b), Belzunce et al. [11] considered $R_{0,n}$ as

$$R_{0,n} = \sum_{k=0}^{\infty} Q^k - \sum_{k=n}^{\infty} Q^k = R - R_n = \sum_{k=0}^{n-1} Q^k,$$

whereas in the proof of Theorem 2(b), $R_{0,n}$ is taken as $R_{0,n} = R - R_{n+1}$. To avoid any confusion, in this paper we stick to the first notation, that is,

$$R_{0,n}(i, j) = \sum_{k=0}^{n-1} Q^k(i, j).$$

For the continuous case, the survival function of L_j is given by

$$\mathbb{P}(L_j > t \mid Y_0 = i) = \mathbb{P}(Y_t \leq j \mid Y_0 = i) = \bar{P}_t(i, j) \quad \text{for all } t \geq 0, i, j \in E$$

if Y starts at i . Now the analogous notations for the process Y are as follows:

- (a) $U(i, j) = \int_0^\infty P_v(i, j) dv$, for all $i, j \in E$, represents the potential matrix,
- (b) $U_t(i, j) = \int_t^\infty P_v(i, j) dv$, for all $t > 0$ and $i, j \in E$, represents the residual potential matrix,
- (c) $U_{t,y}(i, j) = \int_t^{t+y} P_v(i, j) dv$, for $t, y > 0$ and all $i, j \in E$.

Consider the lifetime of a device which is represented by a non-negative random variable T having a cumulative distribution function F . We now briefly discuss the ageing classes which are the focus of the present article.

The concept of *mean time to failure* (MTTF) function was first introduced by Barlow and Proschan [7] in the context of age-replacement policy where the replacement takes place on failure of the item or at a predetermined time t , whichever occurs first. Let $T_{[t]}$ denote the time to the first in-service failure of an equipment under an age-replacement policy with the age-replacement time t . Then the MTTF function corresponding to F (see [7]) is defined as

$$M_F(t) := \mathbb{E}[T_{[t]}] = \frac{\int_0^t \bar{F}(y) dy}{F(t)} \quad \text{for all } t > 0.$$

It is a standard practice to adopt an age-replacement policy in order to prevent unexpected in-service failures, and the MTTF function has traditionally been used to assess the performance and effectiveness of maintenance policy. Two important classes of life distributions based on the MTTF function are defined below.

Definition 2.1. F is said to belong to the *decreasing (increasing) mean time to failure* (DMTTF (IMTTF)) class if $M_F(t)$ is decreasing (increasing) in t .

Li and Xu [56] introduced another ageing family, namely, *new better (worse) than renewal used in the reversed hazard rate order* (NBRU_{rh} (NWRU_{rh})), and also highlighted its properties. However, it turns out that these classes are identical to the DMTTF and IMTTF classes respectively. Preservation under some basic reliability operations, weak convergence issues, as well as closure under convolution are well explored for these classes (see [39], [49], [50]). Our focus is on deriving sufficient conditions which ensure that first-passage times belong to the above classes.

The position of the DMTTF (IMTTF) class in the hierarchy of ageing classes as given in Klefsjö [48] is as follows:

$$\text{IFR (DFR)} \implies \text{IFRA (DFRA)} \implies \text{DMTTF (IMTTF)} \implies \text{NBUE (NWUE)}.$$

The inactive time (IT) (see Ruiz and Navarro [71]), defined by the random time

$$T_{(t)} := [t - T \mid T \leq t],$$

is the amount of time that has elapsed up to time $t > 0$ after failure occurs. Then the *mean inactive time* (MIT) is defined as

$$m_F(t) := \mathbb{E}[T_{(t)}] = \frac{\int_0^t F(y) dy}{F(t)} \quad \text{for all } t > 0.$$

Due to its widespread application in diverse fields, the function is variously known as the *mean waiting time* (MWT) function (see [28]) and the *mean past life* (MPL) function (see [30]). A related ageing class is defined below.

Definition 2.2. F is said to belong to the *increasing mean inactive time* (IMIT) class if $m_F(t)$ is increasing in t .

Chandra and Roy [21] pointed out that it is not possible for a non-negative random variable to have a decreasing mean inactive time (DMIT). The behaviour of the excess lifetime of a renewal process with IMIT inter-arrivals is dealt with in Li and Xu [55]. Characterizations of the IMIT class are established in [1], and weak convergence issues are discussed in [46].

Similarly to the hazard rate, the *reversed hazard rate* (RHR) function of a random variable T is defined as

$$\mu_F(t) := \frac{f(t)}{F(t)} \quad \text{for all } t > 0,$$

where $f(t)$ is the probability density function of T . It was initially termed the ‘dual failure function’ by Keilson and Sumita [42] in the context of characterizing the uniform ordering under shift and scaling. The RHR function has manifold applications including estimation of the survival function in the presence of left censored data (see [4], [34]) and the study of Fisher information arising from weighted distributions (see [31]). In fact it is known as ‘retro-hazard’ in survival analysis. The function also induces an important ageing class, as pointed out in Block et al. [17].

Definition 2.3. F is said to belong to the *decreasing reversed hazard rate* (DRHR) class if $\mu_F(t)$ is decreasing in t .

Block et al. [17] also demonstrate that non-negative random variables cannot have increasing reversed hazard rates. Subsequently, Chandra and Roy [21] proposed a general definition for this class for which the restriction of absolute continuity is not required.

Definition 2.4. A life distribution function F is said to be DRHR if and only if F is log-concave.

Kundu et al. [53] have shown that the DRHR property of a random variable is always transmitted to the corresponding order statistics. The discrete counterpart of the above result has been proved in a recent article by Alimohammadi et al. [3]. The preservation properties of the DRHR class based on the *generalized order statistics* (GOSs) were investigated in [78]. For a general counting process stopped at a random time independent of the process, Badía [6] has established sufficient conditions under which the process is DRHR. Conditions under which the *sequential order statistics* (SOS) possess the DRHR property are studied in Burkschat and Torrado [19].

We now want to highlight the connections between the above two classes and the fundamental class IFR. This has already been nicely illustrated in Chandra and Roy [21] as follows:

$$f \text{ is log-concave} \implies F \text{ is DRHR} \iff F \text{ is log-concave} \implies F \text{ is IMIT}$$

$$\Downarrow$$

$$F \text{ is IFR} \implies Y = -\log_e T \text{ is DRHR.}$$

If the lifetime of a unit is modelled by a discrete random variable S with distribution function $P_k = P(S \leq k)$ and survival function $\bar{P}_k := 1 - P_k$, then in a similar fashion we have the following discrete versions of the above definitions.

- (a) Li and Xu [56]: S is said to be DMTTF (IMTTF) if $\sum_{i=0}^{k-1} \bar{P}_i/P_k$ is decreasing (increasing) in $k \in \mathbb{Z}^+$.
- (b) Goliforushani and Asadi [30]: S is said to be IMIT if $\sum_{i=0}^{k-1} P_i/P_k$ is increasing in $k \in \mathbb{Z}^+$.
- (c) Badía [6]: S is said to be DRHR if P_{k+s}/P_k is decreasing in $k \in \mathbb{Z}^+$ for every $s > 0$.

It thus becomes abundantly clear that the classes considered in the paper have an important place in reliability literature. The authors feel that obtaining sufficient conditions for first-passage times to belong to the above classes would be a significant contribution to the literature.

3. First-passage times for discrete-time Markov chains

Let $X = \{X_n, n \geq 0\}$ be a discrete-time increasing Markov chain with $\{X_0 = i\}$ and transition matrix Q . Consider S_j as defined in (1.1). In this section we explore suitable sufficient conditions for which S_j possesses the above-mentioned ageing criteria. We start out by introducing the following notations:

- (a) $\bar{Q}^n(i, j) := 1 - \bar{Q}^n(i, j)$,
- (b) $\bar{R}_{0,n}(i, j) := \sum_{k=0}^{n-1} \bar{Q}^k(i, j)$.

Note that $\bar{Q}^n(i, j)$ is simply the distribution function of S_j . We call $\bar{R}_{0,n}$ the complement of $\bar{R}_{0,n} = \sum_{k=0}^{n-1} \bar{Q}^k$. We are now in a position to state the main results.

Theorem 3.1. *If $\bar{R}_{0,n}(i, j)/\bar{Q}^n(i, j)$ is decreasing in i for all $n > 0$ and $i, j \in E$, then S_j is DMTTF.*

Proof. By definition, S_j is DMTTF if, for all $0 < n_1 \leq n_2$,

$$\bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \geq \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j).$$

Letting

$$\Theta = \bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) - \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j), \quad (3.1)$$

we only need to show that $\Theta \geq 0$ in order to prove that S_j is DMTTF. Now let $n_2 = n_1 + m$ for some $m \geq 0$; then, using the Chapman–Kolmogorov equations, one can write

$$\begin{aligned} \bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) &= \left(1 - \sum_{l \leq l \leq j} Q^m(i, l) \bar{Q}^{n_1}(l, j)\right) \bar{R}_{0,n_1}(i, j) \\ &= \sum_{l \leq l \leq j} Q^m(i, l) [1 - \bar{Q}^{n_1}(l, j)] \bar{R}_{0,n_1}(i, j) + \sum_{l=j+1}^{\infty} Q^m(i, l) \bar{R}_{0,n_1}(i, j) \end{aligned}$$

and

$$\begin{aligned}\sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) &= \sum_{r=0}^{n_2-1} \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^{r-m}(l, j) \\ &= \sum_{i \leq l \leq j} Q^m(i, l) \sum_{r=0}^{n_2-1} \bar{Q}^{r-m}(l, j) \\ &= \sum_{i \leq l \leq j} Q^m(i, l) \sum_{r=0}^{n_1-1} \bar{Q}^r(l, j).\end{aligned}$$

Then

$$\begin{aligned}\Theta &= \sum_{i \leq l \leq j} Q^m(i, l) [1 - \bar{Q}^{n_1}(l, j)] \bar{R}_{0, n_1}(i, j) - [1 - \bar{Q}^{n_1}(i, j)] \sum_{i \leq l \leq j} Q^m(i, l) \bar{R}_{0, n_1}(l, j) \\ &\quad + \sum_{l=j+1}^{\infty} Q^m(i, l) \bar{R}_{0, n_1}(i, j) \\ &\geq \sum_{i \leq l \leq j} Q^m(i, l) ([1 - \bar{Q}^{n_1}(l, j)] \bar{R}_{0, n_1}(i, j) - [1 - \bar{Q}^{n_1}(i, j)] \bar{R}_{0, n_1}(l, j)).\end{aligned}$$

Therefore $\Theta \geq 0$, that is, S_j is DMTTF if, for all $i \leq l \leq j$,

$$[1 - \bar{Q}^{n_1}(l, j)] \bar{R}_{0, n_1}(i, j) \geq [1 - \bar{Q}^{n_1}(i, j)] \bar{R}_{0, n_1}(l, j),$$

which is equivalent to $\bar{R}_{0, n}(i, j) / \bar{Q}^n(i, j)$ being a decreasing function of i , for all $n > 0$ and $i, j \in E$. $\square$

The mechanism of the above proof does not allow us to obtain the sufficient condition for the dual class IMTTF by just reversing the sense of monotonicity of the function in Theorem 3.1. Although it is apparent that this should be the case, the present authors are unable to either prove or disprove the fact. Instead, the sufficient condition for the corresponding dual class is framed in terms of a new function, as stated in the following theorem.

Theorem 3.2. *If $\bar{R}_{0, n}(i, j) / \bar{Q}^n(i, j)$ is decreasing in i for all $n > 0$ and $i, j \in E$, then S_j is IMTTF.*

Proof. To prove that S_j is IMTTF, we need to show that for all $0 < n_1 \leq n_2$,

$$\bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \leq \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j).$$

Equivalently, if we consider Θ as in (3.1), then in this case it would be sufficient to prove $\Theta \leq 0$. Now, since $n_2 = n_1 + m$ for some $m \geq 0$, we have

$$\begin{aligned}\bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) &= \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) - \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \bar{Q}^{n_1+m}(i, j) \\ &= \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) - \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^{n_1}(l, j)\end{aligned}$$

and

$$\begin{aligned}\bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) &= \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) - \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1+m-1} \bar{Q}^r(i, j) \\ &= \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) - \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^{r+m}(i, j) \\ &= \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) - \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1-1} \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^r(l, j).\end{aligned}$$

Then

$$\begin{aligned}\Theta &= \sum_{i \leq l \leq j} Q^m(i, l) \left(\bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(l, j) - \bar{Q}^{n_1}(l, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \right) - \sum_{r=n_1}^{n_2-1} \bar{Q}^r(i, j) \\ &\leq \sum_{i \leq l \leq j} Q^m(i, l) \left(\bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(l, j) - \bar{Q}^{n_1}(l, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \right).\end{aligned}$$

Thus S_j is IMTTF whenever

$$\bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(l, j) \leq \bar{Q}^{n_1}(l, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) \quad \text{for all } i \leq l \leq j,$$

which follows from the given hypothesis. $\square$

Theorem 3.3. If $\bar{R}_{0,n}(i, j)/\bar{Q}^n(i, j)$ is increasing in i , for all $n > 0$ and $i, j \in E$, then S_j is IMIT.

Proof. Define

$$\Theta = \bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) - \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j).$$

To prove S_j is IMIT we only need to find out such conditions under which $\Theta \leq 0$. Now, using the fact that $n_2 = n_1 + m$, for some $m \geq 0$, and then applying the Chapman–Kolmogorov equations, we have

$$\begin{aligned}\bar{Q}^{n_2}(i, j) \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) &= [1 - \bar{Q}^{n_1+m}(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] \\ &= \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] \left(\sum_{i \leq l \leq j} Q^m(i, l) + \sum_{j+1}^{\infty} Q^m(i, l) - \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^{n_1}(l, j) \right) \\ &= \sum_{i \leq l \leq j} Q^m(i, l) [1 - \bar{Q}^{n_1}(l, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] + \sum_{j+1}^{\infty} Q^m(i, l) \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)]\end{aligned}$$

and also

$$\begin{aligned}
 & \bar{Q}^{n_1}(i, j) \sum_{r=0}^{n_2-1} \bar{Q}^r(i, j) \\
 &= [1 - \bar{Q}^{n_1}(i, j)] \left[(n_2 - 1) - \sum_{r=0}^{n_1-1} \bar{Q}^{r+m}(i, j) \right] \\
 &= [1 - \bar{Q}^{n_1}(i, j)] \left[(n_2 - 1) - \sum_{r=0}^{n_1-1} \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^r(l, j) \right] \\
 &= [1 - \bar{Q}^{n_1}(i, j)] \left((n_2 - 1) + \sum_{i \leq l \leq j} Q^m(i, l) \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(l, j)] - (n_1 - 1) \sum_{i \leq l \leq j} Q^m(i, l) \right) \\
 &= \sum_{i \leq l \leq j} Q^m(i, l) [1 - \bar{Q}^{n_1}(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(l, j)] \\
 &\quad + [1 - \bar{Q}^{n_1}(i, j)] [(n_2 - 1) - (n_1 - 1) \bar{Q}^m(i, j)].
 \end{aligned}$$

Therefore

$$\Theta = \sum_{i \leq l \leq j} Q^m(i, l) \left([1 - \bar{Q}^{n_1}(l, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - [1 - \bar{Q}^{n_1}(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(l, j)] \right) + M,$$

where

$$\begin{aligned}
 M &= \sum_{j+1}^{\infty} Q^m(i, l) \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - [1 - \bar{Q}^{n_1}(i, j)] [(n_2 - 1) - (n_1 - 1) \bar{Q}^m(i, j)] \\
 &= [1 - \bar{Q}^m(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - [1 - \bar{Q}^{n_1}(i, j)] (m + (n_1 - 1) [1 - \bar{Q}^m(i, j)]) \\
 &= [1 - \bar{Q}^m(i, j)] \left(\sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - (n_1 - 1) [1 - \bar{Q}^{n_1}(i, j)] \right) - m [1 - \bar{Q}^{n_1}(i, j)] \\
 &= [1 - \bar{Q}^m(i, j)] M_1 - m [1 - \bar{Q}^{n_1}(i, j)]
 \end{aligned}$$

with

$$\begin{aligned}
 M_1 &= \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - (n_1 - 1) [1 - \bar{Q}^{n_1}(i, j)] \\
 &= - \sum_{r=0}^{n_1-1} \bar{Q}^r(i, j) + (n_1 - 1) \bar{Q}^{n_1}(i, j) \\
 &= [-\bar{Q}^1(i, j) + \bar{Q}^{n_1}(i, j)] + [-\bar{Q}^2(i, j) + \bar{Q}^{n_1}(i, j)] + \cdots + [-\bar{Q}^{n_1-1}(i, j) + \bar{Q}^{n_1}(i, j)] \\
 &\leq 0,
 \end{aligned}$$

since the quantity $\bar{Q}^{n_1}(i, j)$ typically denotes the survival function of S_j for all $n > 0$, and thus each quantity in the parentheses is non-positive. Therefore we obtain

$$\Theta \leq \sum_{i \leq l \leq j} Q^m(i, l) \left([1 - \bar{Q}^{n_1}(l, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - [1 - \bar{Q}^{n_1}(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(l, j)] \right).$$

Thus S_j is IMIT whenever

$$[1 - \bar{Q}^{n_1}(l, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(i, j)] - [1 - \bar{Q}^{n_1}(i, j)] \sum_{r=0}^{n_1-1} [1 - \bar{Q}^r(l, j)] \leq 0 \quad \text{for all } i \leq l \leq j. \quad (3.2)$$

Now, substituting

$$\bar{Q}^n(i, j) = 1 - Q^n(i, j) \quad \text{and} \quad \bar{R}_{0,n}(i, j) = \sum_{r=0}^{n-1} \bar{Q}^r(i, j),$$

(3.2) will be equivalent to $\bar{R}_{0,n}(i, j)/\bar{Q}^n(i, j)$ increasing in $i \in E$, for all $n > 0$ and $j \in E$, which is true by hypothesis. $\square$

Let us recall some important definitions before going on to the next result.

Definition 3.1 (Karlin [36]). Let $A, B \subset \mathbb{R}$. A function $K: A \times B \rightarrow \mathbb{R}^+$ is said to be

- (i) *totally positive of order 2* (TP₂) if $K(x_1, y_1)K(x_2, y_2) - K(x_1, y_2)K(x_2, y_1) \geq 0$,
- (ii) *sign reverse rule of order 2* (RR₂) if $K(x_1, y_1)K(x_2, y_2) - K(x_1, y_2)K(x_2, y_1) \leq 0$,

for all $x_1 < x_2 \in A$ and $y_1 < y_2 \in B$.

We can now proceed to the next theorem.

Theorem 3.4. If $\bar{Q}^n(i, j)$ is RR₂ for $n > 0$ and i ($0 \leq i \leq j$), then S_j is DRHR.

Proof. Exploiting the definition of discrete DRHR, we see that S_j is DRHR whenever the quantity $\bar{Q}^{n+m}(i, j)/\bar{Q}^n(i, j)$ is decreasing in $n \in \mathbb{Z}^+$, for every $m > 0$ and $i, j \in E$, or equivalently

$$\frac{\bar{Q}^{n_1+m}(i, j)}{\bar{Q}^{n_1}(i, j)} \geq \frac{\bar{Q}^{n_2+m}(i, j)}{\bar{Q}^{n_2}(i, j)} \quad \text{for all } 0 < n_1 \leq n_2 \text{ and } i, j \in E.$$

Now, to show that S_j is DRHR, it is enough to show that

$$\Theta := \bar{Q}^{n_1+m}(i, j)\bar{Q}^{n_2}(i, j) - \bar{Q}^{n_2+m}(i, j)\bar{Q}^{n_1}(i, j)$$

is non-negative. A little manipulation and an application of the Chapman–Kolmogorov equations yield

$$\bar{Q}^{n_1+m}(i, j)\bar{Q}^{n_2}(i, j) = \sum_{i \leq l \leq j} Q^m(i, l)[1 - \bar{Q}^{n_1}(l, j)][1 - \bar{Q}^{n_2}(i, j)] + \sum_{j+1}^{\infty} Q^m(i, l)[1 - \bar{Q}^{n_2}(i, j)]$$

and

$$\bar{Q}^{n_2+m}(i, j) \bar{Q}^{n_1}(i, j) = \sum_{i \leq l \leq j} Q^m(i, l) [1 - \bar{Q}^{n_2}(l, j)] [1 - \bar{Q}^{n_1}(i, j)] + \sum_{j+1}^{\infty} Q^m(i, l) [1 - \bar{Q}^{n_1}(i, j)].$$

Therefore

$$\Theta = \sum_{i \leq l \leq j} Q^m(i, l) ([1 - \bar{Q}^{n_1}(l, j)] [1 - \bar{Q}^{n_2}(i, j)] - [1 - \bar{Q}^{n_2}(l, j)] [1 - \bar{Q}^{n_1}(i, j)]) + M,$$

where

$$M = \sum_{j+1}^{\infty} Q^m(i, l) ([1 - \bar{Q}^{n_2}(i, j)] - [1 - \bar{Q}^{n_1}(i, j)]) = \sum_{j+1}^{\infty} Q^m(i, l) [\bar{Q}^{n_1}(i, j) - \bar{Q}^{n_2}(i, j)].$$

Since $n_2 = n_1 + k$ for some $k \geq 0$, we have

$$\begin{aligned} M &= \sum_{j+1}^{\infty} Q^m(i, l) [\bar{Q}^{n_1}(i, j) - \bar{Q}^{n_1+k}(i, j)] \\ &= \sum_{j+1}^{\infty} Q^m(i, l) \left[\sum_{r \leq j} Q^{n_1}(i, r) - \sum_{i \leq r \leq j} Q^{n_1}(i, r) \bar{Q}^k(r, j) \right] \\ &= \sum_{j+1}^{\infty} Q^m(i, l) \left[\sum_{r \leq j} Q^{n_1}(i, r) (1 - \bar{Q}^k(r, j)) \right] \\ &\geq 0. \end{aligned}$$

This implies

$$\Theta \geq \sum_{i \leq l \leq j} Q^m(i, l) ([1 - \bar{Q}^{n_1}(l, j)] [1 - \bar{Q}^{n_2}(i, j)] - [1 - \bar{Q}^{n_2}(l, j)] [1 - \bar{Q}^{n_1}(i, j)]) \geq 0$$

whenever

$$\bar{Q}^{n_1}(l, j) \bar{Q}^{n_2}(i, j) - \bar{Q}^{n_2}(l, j) \bar{Q}^{n_1}(i, j) \geq 0,$$

where

$$\bar{Q}^n(i, j) = 1 - \bar{Q}^n(i, j) \geq 0 \quad \text{for all } n \text{ and } i, j \in E.$$

This is equivalent to saying that the function $\bar{Q}^n(i, j)$ is RR_2 for the variables $n > 0$ and i ($0 \leq i \leq j$) where $i, j \in E$. $\square$

4. First-passage times for continuous-time Markov processes

Let $Y = \{Y_t, t \geq 0\}$ be the regular Markov process as discussed in Section 1, with $\{Y_0 = i\}$ and $P_t(i, j)$ being the transition function of the process. Now, as before, we introduce the

following notations:

- (a) $\bar{P}_t(i, j) := 1 - \bar{P}_t(i, j)$,
- (b) $\bar{U}_{0,t}(i, j) := \int_0^t \bar{P}_v(i, j) dv$ for all $t > 0$ and $i, j \in E$,

where $\bar{U}_{0,t}(i, j)$ represents the complement of

$$\bar{U}_{0,t}(i, j) = \int_0^t \bar{P}_v(i, j) dv$$

and $\bar{P}_t(i, j)$ is the distribution function of L_j . Then, under the framework of a continuous-time parameter, we have the following results.

Theorem 4.1. *Let the first-passage time L_j be defined as in (1.2).*

- (i) *If $\bar{U}_{0,t}(i, j)/\bar{P}_t(i, j)$ is decreasing in $i \in E$ for all $t > 0$ and $j \in E$, then L_j is DMTTF.*
- (ii) *If $\bar{U}_{0,t}(i, j)/\bar{P}_t(i, j)$ is decreasing in $i \in E$ for all $t > 0$ and $j \in E$, then L_j is IMTTF.*
- (iii) *If $\bar{U}_{0,t}(i, j)/\bar{P}_t(i, j)$ is increasing in $i \in E$ for all $t > 0$ and $j \in E$, then L_j is IMIT.*
- (iv) *If $\bar{P}_t(i, j)$ is RR_2 for the variables $t > 0$ and i ($0 \leq i \leq j$), then L_j is DRHR.*

Proof. A brief sketch of the proofs of (i), (ii), and (iv) is given. The proof of (iii) needs a little care.

(i) To prove that L_j is DMTTF, we need to show

$$\Phi = \bar{P}_{t_2}(i, j) \int_0^{t_1} \bar{P}_v(i, j) dv - \bar{P}_{t_1}(i, j) \int_0^{t_2} \bar{P}_v(i, j) dv \geq 0 \quad \text{for all } 0 < t_1 \leq t_2. \quad (4.1)$$

Applying the Chapman–Kolmogorov equations, by an argument similar to that in the discrete case, we obtain

$$\begin{aligned} \Phi &= \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{P}_{t_1}(l, j) \bar{U}_{0,t_1}(i, j) + \sum_{l=j+1}^{\infty} P_{t_2-t_1}(i, l) \bar{U}_{0,t_1}(i, j) \\ &\quad - \bar{P}_{t_1}(i, j) \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{U}_{0,t_1}(l, j) \\ &\geq \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) (\bar{P}_{t_1}(l, j) \bar{U}_{0,t_1}(i, j) - \bar{P}_{t_1}(i, j) \bar{U}_{0,t_1}(l, j)). \end{aligned}$$

Therefore $\Phi \geq 0$ whenever

$$\bar{P}_{t_1}(l, j) \bar{U}_{0,t_1}(i, j) \geq \bar{P}_{t_1}(i, j) \bar{U}_{0,t_1}(l, j).$$

Thus the condition ensuring the DMTTF property of L_j can be stated as

$$\frac{\bar{U}_{0,t}(i, j)}{\bar{P}_t(i, j)} \geq \frac{\bar{U}_{0,t}(l, j)}{\bar{P}_t(l, j)} \quad \text{for all } i \leq l \leq j,$$

or equivalently $\bar{U}_{0,t}(i, j)/\bar{P}_t(i, j)$ is decreasing in i , for all $t > 0$ and $i, j \in E$.

(ii) Defining Φ as (4.1) above, we deduce that L_j is IMTTF if $\Phi \leq 0$. Now an analogous manipulation to that in Theorem 3.2 yields

$$\begin{aligned}\Phi &= \int_0^{t_1} \bar{P}_v(i, j) \, dv \left(1 - \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{P}_{t_1}(l, j) \right) - \int_0^{t_2} \bar{P}_v(i, j) \, dv \\ &\quad + \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{P}_{t_1}(i, j) \int_0^{t_1} \bar{P}_v(l, j) \, dv \\ &\leq \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) (\bar{P}_{t_1}(i, j) \bar{U}_{0, t_1}(l, j) - \bar{P}_{t_1}(l, j) \bar{U}_{0, t_1}(i, j)).\end{aligned}$$

Therefore $\Phi \leq 0$ for

$$\frac{\bar{U}_{0, t_1}(l, j)}{\bar{P}_{t_1}(l, j)} \leq \frac{\bar{U}_{0, t_1}(i, j)}{\bar{P}_{t_1}(i, j)} \quad \text{for all } i \leq l \leq j,$$

which is true from the given hypothesis.

(iii) To show that L_j possesses the IMIT property, it suffices to show that

$$\Phi = \bar{P}_{t_2}(i, j) \int_0^{t_1} \bar{P}_v(i, j) \, dv - \bar{P}_{t_1}(i, j) \int_0^{t_2} \bar{P}_v(i, j) \, dv \leq 0.$$

As before,

$$\bar{P}_{t_2}(i, j) \int_0^{t_1} \bar{P}_v(i, j) \, dv = \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{P}_{t_1}(l, j) \bar{U}_{0, t_1}(i, j) + \sum_{l=j+1}^{\infty} P_{t_2-t_1}(i, l) \bar{U}_{0, t_1}(i, j)$$

and

$$\begin{aligned}\bar{P}_{t_1}(i, j) \int_0^{t_2} \bar{P}_v(i, j) \, dv &= \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) \bar{U}_{0, t_1}(l, j) \bar{P}_{t_1}(i, j) + (t_2 - t_1) \bar{P}_{t_2-t_1}(i, j) \bar{P}_{t_1}(i, j).\end{aligned}$$

Thus

$$\Phi = \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) (\bar{P}_{t_1}(l, j) \bar{U}_{0, t_1}(i, j) - \bar{U}_{0, t_1}(l, j) \bar{P}_{t_1}(i, j)) + M', \quad (4.2)$$

where

$$\begin{aligned}M' &= \sum_{l=j+1}^{\infty} P_{t_2-t_1}(i, l) \bar{U}_{0, t_1}(i, j) - \bar{P}_{t_1}(i, j) (t_2 - t_1) \bar{P}_{t_2-t_1}(i, j) \\ &= \bar{P}_{t_2-t_1}(i, j) \bar{U}_{0, t_1}(i, j) - \bar{P}_{t_1}(i, j) ((t_2 - t_1) + t_1 \bar{P}_{t_2-t_1}(i, j)) \\ &= \bar{P}_{t_2-t_1}(i, j) M'_1 - (t_2 - t_1) \bar{P}_{t_1}(i, j),\end{aligned}$$

with

$$M'_1 = \bar{U}_{0,t_1}(i, j) - t_1 \bar{P}_{t_1}(i, j) = - \int_0^{t_1} \bar{P}_v(i, j) \, dv + t_1 \bar{P}_{t_1}(i, j).$$

Note that $\bar{P}_t(i, j)$ is the survival function of L_j and hence is a decreasing function of t . Therefore we have

$$\begin{aligned} \bar{P}_{t_1}(i, j) \leq \bar{P}_v(i, j) \quad \text{for all } v \leq t_1 &\implies \int_0^{t_1} \bar{P}_{t_1}(i, j) \, dv \leq \int_0^{t_1} \bar{P}_v(i, j) \, dv \\ &\implies t_1 \bar{P}_{t_1}(i, j) \leq \int_0^{t_1} \bar{P}_v(i, j) \, dv \\ &\implies M'_1 \leq 0 \end{aligned}$$

and thus $M' \leq 0$. Now (4.2) implies

$$\Phi \leq \sum_{i \leq l \leq j} P_{t_2-t_1}(i, l) (\bar{P}_{t_1}(l, j) \bar{U}_{0,t_1}(i, j) - \bar{U}_{0,t_1}(l, j) \bar{P}_{t_1}(i, j)).$$

Therefore L_j enjoys the IMIT property when

$$\frac{\bar{U}_{0,t_1}(i, j)}{\bar{P}_{t_1}(i, j)} \leq \frac{\bar{U}_{0,t_1}(l, j)}{\bar{P}_{t_1}(l, j)} \quad \text{for all } i \leq l \leq j$$

or, in other words, $\bar{U}_{0,t}(i, j)/\bar{P}_t(i, j)$ is increasing in $i \in E$, for all $t \geq 0$ and $j \in E$.

(iv) From Definition 2.4, L_j is DRHR if and only if $\bar{P}_t(i, j)$, the distribution function of L_j , is log-concave, that is, for all $0 < t_1 \leq t_2$ and for every $s > 0$,

$$\frac{\bar{P}_{t_1+s}(i, j)}{\bar{P}_{t_1}(i, j)} \geq \frac{\bar{P}_{t_2+s}(i, j)}{\bar{P}_{t_2}(i, j)}.$$

that is,

$$\bar{P}_{t_1+s}(i, j) \bar{P}_{t_2}(i, j) - \bar{P}_{t_2+s}(i, j) \bar{P}_{t_1}(i, j) \geq 0 \quad \text{for all } 0 < t_1 \leq t_2 \text{ and } s > 0.$$

Defining Φ as the expression on the left-hand side of the above inequality, we have by applying the Chapman–Kolmogorov equations

$$\Phi = \sum_{i \leq l \leq j} P_s(i, l) (\bar{P}_{t_1}(l, j) \bar{P}_{t_2}(i, j) - \bar{P}_{t_2}(l, j) \bar{P}_{t_1}(i, j)) + M',$$

where

$$\begin{aligned} M' &= \sum_{l=j+1}^{\infty} P_s(i, l) (\bar{P}_{t_2}(i, j) - \bar{P}_{t_1}(i, j)) \\ &= \sum_{l=j+1}^{\infty} P_s(i, l) (\bar{P}_{t_1}(i, j) - \bar{P}_{t_2}(i, j)) \geq 0 \quad \text{for all } 0 < t_1 \leq t_2 \end{aligned}$$

as $\bar{P}_t(i, j)$, being the survival function, is decreasing in t . Hence

$$\Phi \geq \sum_{i \leq l \leq j} P_s(i, l) (\bar{P}_{t_1}(l, j) \bar{P}_{t_2}(i, j) - \bar{P}_{t_2}(l, j) \bar{P}_{t_1}(i, j)).$$

Therefore $\Phi \geq 0$, or equivalently L_j is DRHR if

$$\bar{P}_{t_1}(l, j) \bar{P}_{t_2}(i, j) - \bar{P}_{t_2}(l, j) \bar{P}_{t_1}(i, j) \geq 0. \quad (4.3)$$

Now condition (4.3) holds due to the fact that $\bar{P}_t(i, j)$ is RR_2 for the variables $t > 0$ and i ($0 \leq i \leq j$), where $i, j \in E$. $\square$

5. Discussion

The present work is largely concerned with obtaining appropriate conditions in the presence of which first-passage times of an increasing Markov process enjoy the DMTTF (IMTTF), IMIT, and DRHR properties. At this juncture it is pertinent to add some comments. Unlike in the case of the classical reliability classes, the functions giving rise to the ageing classes considered here have some specific forms where the distribution function figures prominently rather than the survival function. This aspect poses significant challenges in establishing relevant results concerning the first-passage times. The proofs involved for obtaining the appropriate conditions for S_j and L_j are neither routine nor straightforward. Also, typically the sufficient condition for the dual class can be obtained by just reversing the sense of monotonicity of the function which is present in the sufficient condition for the original class (see e.g. [11]). This, however, is not the case for the DMTTF and IMTTF classes (see Theorems 3.1 and 3.2 for details). It is possible that such features appear due to the specific structure of the underlying reliability function. Again, observe that the mechanism of proof utilized in Theorem 3.4 of the present article allows us to provide a sufficient condition for S_j to be IFR (DFR). Before moving on to this, we first recapitulate the following definition.

Definition 5.1 (Barlow and Proschan [7]). A life distribution function F is said to be IFR (DFR) if and only if $\bar{F}$ is log-concave (log-convex).

Theorem 5.1. Let the first-passage time S_j of a discrete time increasing Markov chain X with transition matrix Q be defined as in Section 1.

- (i) If $\bar{Q}^n(i, j)$ is RR_2 for the variables $n > 0$ and i ($0 \leq i \leq j$), then S_j is IFR.
- (ii) If $\bar{Q}^n(i, j)$ is TP_2 for the variables $n > 0$ and i ($0 \leq i \leq j$), then S_j is DFR.

Proof. (i) From Definition 5.1, S_j is IFR if and only if $\bar{Q}^n(i, j)$ is log-concave, or equivalently $\bar{Q}^{n+m}(i, j)/\bar{Q}^n(i, j)$ is decreasing in $n > 0$ for every $m > 0$ and $i, j \in E$, that is,

$$\frac{\bar{Q}^{n_1+m}(i, j)}{\bar{Q}^{n_1}(i, j)} \geq \frac{\bar{Q}^{n_2+m}(i, j)}{\bar{Q}^{n_2}(i, j)} \quad \text{for all } 0 < n_1 \leq n_2.$$

Now let

$$\Theta = \bar{Q}^{n_1+m}(i, j) \bar{Q}^{n_2}(i, j) - \bar{Q}^{n_2+m}(i, j) \bar{Q}^{n_1}(i, j).$$

Applying the Chapman–Kolmogorov equations, we obtain

$$\begin{aligned}\Theta &= \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^{n_1}(l, j) \bar{Q}^{n_2}(i, j) - \sum_{i \leq l \leq j} Q^m(i, l) \bar{Q}^{n_2}(l, j) \bar{Q}^{n_1}(i, j) \\ &= \sum_{i \leq l \leq j} Q^m(i, l) (\bar{Q}^{n_1}(l, j) \bar{Q}^{n_2}(i, j) - \bar{Q}^{n_1}(i, j) \bar{Q}^{n_2}(l, j)).\end{aligned}$$

Therefore $\Theta \geq 0$, that is, S_j is IFR whenever $\bar{Q}^{n_1}(l, j) \bar{Q}^{n_2}(i, j) - \bar{Q}^{n_1}(i, j) \bar{Q}^{n_2}(l, j) \geq 0$. The proof now follows by noting that this inequality is immediate from the given hypothesis.

(ii) The proof is similar and hence omitted for the sake of brevity. $\square$

Theorem 5.1 can be extended to the continuous case as well.

Theorem 5.2. *Let the first-passage time L_j of an increasing Markov process Y with continuous-time parameter and having transition function P_t be defined as in Section 1.*

- (i) *If $\bar{P}_t(i, j)$ is RR_2 for the variables $t > 0$ and i ($0 \leq i \leq j$), then L_j is IFR.*
- (ii) *If $\bar{P}_t(i, j)$ is TP_2 for the variables $t > 0$ and i ($0 \leq i \leq j$), then L_j is DFR.*

Proof. The proof is analogous to that of Theorem 5.1 and hence omitted. $\square$

It is worth noting that the hierarchical order found in the ageing classes can also be observed in the corresponding sufficient conditions. For instance, Brown and Chaganty [18] and Shaked and Shanthikumar [72] have shown that the sufficient condition for the first-passage time to have the IFR property is that the transition matrix $Q(i, j)$ is TP_2 , while for the IFRA case, it turns out that $\bar{Q}(i, j)$ is decreasing in i for all j . By using Remark 1.1 of [40], it is easy to see that $Q(i, j)$ being TP_2 implies that $\bar{Q}(i, j)$ is decreasing in i for all j . Again, Karasu and Özekici [35] pointed out that the condition that $\bar{Q}(i, j)$ is decreasing in i for all j further implies that $\bar{R}(i, j)$ is decreasing in i for all j , which in turn indicates that the first-passage time possesses the NBUE property. Pérez-Ocón and Gámiz-Pérez [67] have established that if $\bar{R}_n(i, j)/\bar{Q}^n(i, j)$ is decreasing in i for all $n > 0$ and $i, j \in E$, then S_j is DMRL. Simple algebra shows that the sufficient condition obtained in Theorem 5.1 for the IFR class implies that $\bar{R}_n(i, j)/\bar{Q}^n(i, j)$ is decreasing in i for all $n > 0$ and $i, j \in E$. While establishing such connections between the conditions of IFR (as given in Brown and Chaganty [18]) and IFR(2) classes (given in [11]), Belzunce et al. [11] have come up with some incorrect statements. In this connection, it is also important to note that for a continuous life distribution, the classes IFR and IFR(2) are equivalent (see Theorem 3.8 in [10] or Proposition 2.3 in [9]). Although, in [11], these incorrect statements arise in the context of IFR and IFR(2) ageing notions, we observe that they are fundamentally related to the TP_2 and RR_2 properties of a function (see Table 1). Thus the present authors find it pertinent to investigate this issue. Throughout this discussion, we adopt the notation of [11] for convenience.

First of all, note that if the function $\bar{K}^k(i, j)$ is TP_2 in (k, i) , then simple algebra yields

$$\sum_{k=n}^{n+m} \bar{K}^k(i, j) \bar{K}^n(r, j) \leq \sum_{k=n}^{n+m} \bar{K}^k(r, j) \bar{K}^n(i, j) \quad \text{for all } i \leq r \leq j \text{ and } n, m \in \mathbb{Z}^+.$$

Thus the statement in [11, p. 255] that ‘if $\bar{K}^k(i, j)$ is TP_2 in (k, i) , then (18) holds’, where (18) is the inequality given by ‘ $\sum_{k=n}^{n+m} \bar{K}^k(i, j) \bar{K}^n(r, j) \geq \sum_{k=n}^{n+m} \bar{K}^k(r, j) \bar{K}^n(i, j)$ for all $i \leq r \leq j$

TABLE 1. Summary of errors in [11] and their corresponding correct versions.

Erroneous statements in [11]	Correct implications
$\bar{K}^k(i, j)$ is TP ₂ in $(k, i) \implies$ $\sum_{k=n}^{n+m} \bar{K}^k(i, j) \bar{K}^n(r, j) \geq \sum_{k=n}^{n+m} \bar{K}^k(r, j) \bar{K}^n(i, j),$ for all $i \leq r \leq j$ and $n, m \in \mathbb{Z}^+$	The inequality should be reversed
$K(i, j)$ is TP ₂ in $(i, j) \implies \bar{K}^k(i, j)$ is TP ₂ in (k, i)	$\bar{K}^k(i, j)$ is RR ₂ in (k, i)
$K^k(i, j)$ is TP ₂ in $(i, j) \implies \bar{K}^k(i, j)$ is TP ₂ in (k, i)	$\bar{K}^k(i, j)$ is TP ₂ in (i, j)
$K(i, j)$ is TP ₂ in $(i, j) \implies K^k(i, j)$ is TP ₂ in (k, i)	$K^k(i, j)$ is RR ₂ in (k, i)

and $n, m \in \mathbb{Z}^+$, is incorrect. In fact, the inequality should be reversed. Consequently, from the condition of Theorem 1(a) in [11], we have that if $\bar{K}^k(i, j)$ is TP₂ in (k, i) then S_j is DFR(2). Again, from [18], S_j is IFR whenever $K(i, j)$ is TP₂ in (i, j) . This observation forces us to conclude that Proposition 1 of [11] is incorrect. If not, it will follow that if the first-passage time S_j is IFR then it will be DFR(2) as well. In this connection we note that in the proof of Proposition 1, Belzunce et al. [11] have claimed, by referring to Corollary 3.1 of Kijima [47, p. 108], that in order to show that $\bar{K}^k(i, j)$ is TP₂ in (k, i) , it would be enough to prove that $K^k(i, j)$ is TP₂ in (k, i) . However, using Corollary 3.1 of [47], it can only be shown that $K^k(i, j)$ being TP₂ for (i, j) implies that $\bar{K}^k(i, j)$ is TP₂ for (i, j) , and not for the variables (k, i) , as claimed in [11]. Moreover, exploiting Theorem 2.4 in Karlin [36] for the case $r = 2$, one obtains that $\bar{K}^k(i, j)$ is in fact RR₂ in (k, i) whenever $K(i, j)$ is TP₂ in (i, j) . This contradicts the statement of Proposition 1 in [11].

Again, in the proof of Proposition 1, Belzunce et al. [11] have tried to establish that $K^k(i, j)$ is TP₂ in (k, i) whenever the condition $K(i, j)$ is TP₂ in (i, j) holds, by using a result pertaining to stochastic monotonicity. Belzunce et al. [11] claimed that since the TP₂ property implies stochastic monotonicity (see [41]), therefore the condition ' $K(r_2, j) \geq K(r_1, j)$ for $r_1 \leq r_2$ ' holds due to the stochastic monotonicity property of the transition matrix K . However, this is not the case, as the notion of stochastic monotonicity deals with the row vectors of a matrix and not with some particular entries of it (see Definition 1.2 of Keilson and Kester [41] for details). In this connection, note from (2.12) of Theorem 2.4 of Karlin [36] that $K(i, j)$ being TP₂ in (i, j) implies the condition that $K^k(i, j)$ is RR₂ in (k, i) .

At this juncture it seems appropriate to summarize the above discussion in a structured way. To this end, we present Table 1.

Next we highlight some connections among the sufficient conditions in the context of the reliability classes discussed in this paper. From Brown and Chaganty [18], once again note that S_j is IFRA whenever $\bar{Q}(i, j)$ is decreasing in i for all $i, j \in E$. This condition implies that for all $i, j \in E$, $\bar{R}_{0,n}(i, j) = \sum_{k=0}^{n-1} \bar{Q}(i, j)$ is decreasing in i and also $\bar{Q}^n(i, j) = 1 - \bar{Q}^n(i, j)$ is increasing in i . This in turn leads to the conclusion that $\bar{R}_{0,n}(i, j)/\bar{Q}^n(i, j)$ is decreasing in i for all $n > 0$ and $i, j \in E$, which is a sufficient condition for S_j to be DMTTF. On the other hand, to show that the sufficient condition for the case of the DRHR class implies that of the IMIT class, first we need to observe that the condition that $\bar{Q}^n(i, j)$ is RR₂ for $n > 0$ and i ($0 \leq i \leq j$) is equivalent to

$$\frac{\bar{Q}^{n_1}(i, j)}{\bar{Q}^{n_2}(i, j)} \leq \frac{\bar{Q}^{n_1}(r, j)}{\bar{Q}^{n_2}(r, j)} \quad \text{for all } i \leq r \leq j \text{ and } 0 < n_1 \leq n_2. \quad (5.1)$$

Here our goal is to show that in the presence of condition (5.1), the function $\bar{R}_{0,n}(i, j)/\bar{Q}^n(i, j)$ as given in Theorem 3.3 is increasing in i for all $n > 0$ and $i, j \in E$. Now using (5.1) we have, for all $i \leq r \leq j$,

$$\begin{aligned} \frac{\bar{R}_{0,n}(i, j)}{\bar{Q}^n(i, j)} &= \frac{\sum_{k=0}^{n-1} \bar{Q}^k(i, j)}{\bar{Q}^n(i, j)} \\ &= \frac{\bar{Q}^1(i, j)}{\bar{Q}^n(i, j)} + \frac{\bar{Q}^2(i, j)}{\bar{Q}^n(i, j)} + \cdots + \frac{\bar{Q}^{n-1}(i, j)}{\bar{Q}^n(i, j)} \\ &\leq \frac{\bar{Q}^1(r, j)}{\bar{Q}^n(r, j)} + \frac{\bar{Q}^2(r, j)}{\bar{Q}^n(r, j)} + \cdots + \frac{\bar{Q}^{n-1}(r, j)}{\bar{Q}^n(r, j)} \\ &= \frac{\sum_{k=0}^{n-1} \bar{Q}^k(r, j)}{\bar{Q}^n(r, j)} \\ &= \frac{\bar{R}_{0,n}(r, j)}{\bar{Q}^n(r, j)}, \end{aligned}$$

which completes the proof.

Finally, we end this section by adding some examples to strengthen the practical impact of the work. Our first example presents a Markov chain whose transition matrix satisfies the condition of Theorem 3.2 so that the first-passage time possesses the IMTTF property. It is natural that (except in the trivial case where the distribution of the first-passage time is exponential) if S_j satisfies the IMTTF property then it cannot satisfy the condition for its dual class DM TTF. The following example also demonstrates this fact.

Example 5.1. Consider a discrete-time increasing Markov chain X with transition matrix Q given by

$$Q = \frac{1}{5} \begin{pmatrix} 1 & 1 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 5 \end{pmatrix}.$$

By induction it can be verified that

$$Q^n = \frac{1}{5^n} \begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) & 5^n - \frac{1}{2}(3^n + 1) \\ 0 & 3^n & 5^n - 3^n \\ 0 & 0 & 5^n \end{pmatrix}.$$

It is now easy to see from Theorem 3.2 that S_j is IMTTF whenever

$$\bar{Q}^n(l, j) \sum_{r=0}^{n-1} \bar{Q}^r(i, j) \geq \bar{Q}^n(i, j) \sum_{r=0}^{n-1} \bar{Q}^r(l, j) \quad (5.2)$$

holds for all $i \leq l \leq j$. Taking $i = 1, l = 2, j = 2$, we have from (5.2)

$$\begin{aligned} \text{LHS} &= \bar{Q}^n(2, 2) \sum_{r=0}^{n-1} \bar{Q}^r(1, 2) \\ &= \left(\frac{3}{5}\right)^n \sum_{r=0}^{n-1} \frac{1}{2 \times 5^r} (3^r + 1) \\ &= \frac{1}{2} \left[\left(\frac{3}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r + \left(\frac{3}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{1}{5}\right)^r \right] \end{aligned}$$

and

$$\begin{aligned} \text{RHS} &= \bar{Q}^n(1, 2) \sum_{r=0}^{n-1} \bar{Q}^r(2, 2) \\ &= \frac{1}{2 \times 5^n} (3^n + 1) \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r \\ &= \frac{1}{2} \left[\left(\frac{3}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r + \left(\frac{1}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r \right]. \end{aligned}$$

Now

$$\begin{aligned} 2 \text{ LHS} - 2 \text{ RHS} &= \left(\frac{3}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{1}{5}\right)^r - \left(\frac{1}{5}\right)^n \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r \\ &= \frac{1}{5^n} \left[3^n \left(1 + \frac{1}{5} + \frac{1}{5^2} + \cdots + \frac{1}{5^{n-1}}\right) - \left(1 + \frac{3}{5} + \frac{3^2}{5^2} + \cdots + \frac{3^{n-1}}{5^{n-1}}\right) \right] \\ &> 0 \quad \text{for all } n > 0, \end{aligned}$$

that is, the condition (5.2) is satisfied for the choice $i = 1, l = 2, j = 2$. Note that for all other possible choices of i, l , and j , (5.2) is trivially satisfied. Thus the first-passage time S_j possesses the IMTTF property.

Now we show that S_j does not satisfy the condition of Theorem 3.1. The condition of Theorem 3.1 can be written as follows:

$$[1 - \bar{Q}^n(l, j)] \sum_{r=0}^{n-1} \bar{Q}^r(i, j) \geq [1 - \bar{Q}^n(i, j)] \sum_{r=0}^{n-1} \bar{Q}^r(l, j) \quad \text{for all } i \leq l \leq j. \quad (5.3)$$

Taking $i = 1, l = 2, j = 2$, we have from (5.3)

$$\begin{aligned} \text{LHS} &= [1 - \bar{Q}^n(2, 2)] \sum_{r=0}^{n-1} \bar{Q}^r(1, 2) \\ &= \left[1 - \left(\frac{3}{5}\right)^n\right] \sum_{r=0}^{n-1} \frac{1}{2 \times 5^r} (3^r + 1) \\ &= \frac{1}{2} \left[\left(1 - \left(\frac{3}{5}\right)^n\right) \sum_{r=0}^{n-1} \left(\frac{3}{5}\right)^r + \left(1 - \left(\frac{3}{5}\right)^n\right) \sum_{r=0}^{n-1} \left(\frac{1}{5}\right)^r \right] \end{aligned}$$

and

$$\begin{aligned} \text{RHS} &= [1 - \bar{Q}^n(1, 2)] \sum_{r=0}^{n-1} \bar{Q}^r(2, 2) \\ &= \left[1 - \frac{1}{2 \times 5^n} (3^n + 1) \right] \sum_{r=0}^{n-1} \left(\frac{3}{5} \right)^r \\ &= \frac{1}{2} \left[\left(1 - \left(\frac{3}{5} \right)^n \right) \sum_{r=0}^{n-1} \left(\frac{3}{5} \right)^r + \left(1 - \left(\frac{1}{5} \right)^n \right) \sum_{r=0}^{n-1} \left(\frac{3}{5} \right)^r \right]. \end{aligned}$$

From the above expressions it is easy to observe that $\text{LHS} < \text{RHS}$ for that particular choice of i , l , and j . This clearly violates the condition (5.3).

The next example arises in a study of breast cancer. Pérez-Ocón et al. [68] considered a continuous-time homogeneous Markov process to model the effect of treatments in survival of breast cancer. A cohort of 300 patients were given a combination of three different treatments such as chemotherapy (CT), radiotherapy (RT), and hormonal therapy (HT). In each case the impact of those treatments was categorized into three states, namely no relapse (state 1), relapse (state 2), and death (state 3). Belzunce et al. [11] considered the case where all three treatments were applied to the patients, and subsequently investigated the transition matrix obtained in this scenario. Taking a cue from [11], here we study the same model and show that the first-passage time of the Markov process in this case possesses the DMTTF ageing property as well.

Example 5.2. Consider the continuous-time homogeneous Markov process that models the effects of various treatments in the study of breast cancer (as discussed above) with the transition function given by

$$P_t = \begin{pmatrix} e^{-(q_{12}+q_{13})t} & \frac{q_{12}(e^{-(q_{12}+q_{13})t} - e^{q_{23}t})}{q_{23} - q_{12} - q_{13}} & 1 - p_{11}(t) - p_{12}(t) \\ 0 & e^{-q_{13}t} & 1 - e^{-q_{13}t} \\ 0 & 0 & 1 \end{pmatrix},$$

where q_{ij} is the transition intensity between the states i, j of the Markov process whose estimated values are given by (see [11])

$$\hat{q}_{12} = 0.0080, \quad \hat{q}_{13} = 0.0024, \quad \hat{q}_{23} = 0.0408.$$

Now, to check the DMTTF property of L_j , observe that condition (i) of Theorem 4.1 can be expressed as

$$\bar{P}_t(l, j) \bar{U}_{0,t}(i, j) \geq \bar{P}_t(i, j) \bar{U}_{0,t}(l, j) \quad \text{for all } i \leq l \leq j. \quad (5.4)$$

Take $i = 1$, $l = 2$, and $j = 2$. Writing $\hat{q}_{12} = a$, $\hat{q}_{13} = b$, and $\hat{q}_{23} = c$, from (5.4) we have

$$\begin{aligned} \text{LHS} &= \bar{P}_t(2, 2) \bar{U}_{0,t}(1, 2) \\ &= (1 - e^{-ct}) \int_0^t \left(\frac{c-b}{c-a-b} e^{-(a+b)u} - \frac{a}{c-a-b} e^{-cu} \right) du \\ &= (1 - e^{-ct}) \left[\frac{(c-b)}{(c-a-b)(a+b)} (1 - e^{-(a+b)t}) - \frac{a}{c(c-a-b)} (1 - e^{-ct}) \right] \end{aligned}$$

and

$$\begin{aligned} \text{RHS} &= \bar{P}_t(1, 2)\bar{U}_{0,t}(2, 2) \\ &= \left(1 - \frac{c-b}{c-a-b}e^{-(a+b)t} + \frac{a}{c-a-b}e^{-ct}\right) \int_0^t e^{-cu} du \\ &= \left(1 - \frac{c-b}{c-a-b}e^{-(a+b)t} + \frac{a}{c-a-b}e^{-ct}\right) \left(\frac{1}{c} - \frac{e^{-ct}}{c}\right). \end{aligned}$$

After some algebraic manipulation, we obtain

$$\text{LHS} - \text{RHS} = \frac{c-b}{c(a+b)}(1 - e^{-ct})(1 - e^{-(a+b)t}) \geq 0, \quad \text{since } c > b.$$

Thus the condition of (5.4) is satisfied for $i = 1$, $l = 2$, and $j = 2$. For the other choices, the condition trivially holds. Hence L_j is DMTTF.

We complete our work with the following example, where the conditions for the DRHR and the IMIT classes are both satisfied.

Example 5.3. Consider a continuous-time homogeneous Markov process with transition function defined by

$$P_t = \begin{pmatrix} \frac{1}{4} & \frac{1}{4}e^{-4t} & \frac{3}{4} - \frac{1}{4}e^{-4t} \\ 0 & \frac{1}{4} - \frac{1}{4}e^{-4t} & \frac{3}{4} + \frac{1}{4}e^{-4t} \\ 0 & 0 & 1 \end{pmatrix}.$$

The condition in (iv) of Theorem 4.1 is equivalent to

$$\bar{P}_{t_1}(i, j)\bar{P}_{t_2}(l, j) \leq \bar{P}_{t_1}(l, j)\bar{P}_{t_2}(i, j) \quad (5.5)$$

for all $t_1 < t_2$ and $i \leq l \leq j$. Thus, for $t_1 < t_2$, one only needs to verify the case where $i = 1$, $l = 2$, and $j = 2$. Now

$$\begin{aligned} \bar{P}_{t_1}(1, 2)\bar{P}_{t_2}(2, 2) &= \left[1 - \left(\frac{1}{4} + \frac{1}{4}e^{-4t_1}\right)\right] \left[1 - \left(\frac{1}{4} - \frac{1}{4}e^{-4t_2}\right)\right] \\ &= \frac{1}{16}[9 - 3e^{-4t_1} + 3e^{-4t_2} - e^{-4(t_1+t_2)}] \end{aligned}$$

and

$$\bar{P}_{t_1}(2, 2)\bar{P}_{t_2}(1, 2) = \frac{1}{16}[9 + 3e^{-4t_1} - 3e^{-4t_2} - e^{-4(t_1+t_2)}].$$

Now it is easy to see that (5.5) is satisfied and therefore L_j is DRHR.

Again, to show that L_j possesses the IMIT property, note that condition (iii) of Theorem 4.1 can be expressed as

$$\bar{P}_t(l, j)\bar{U}_{0,t}(i, j) \leq \bar{U}_{0,t}(l, j)\bar{P}_t(i, j), \quad i \leq l \leq j. \quad (5.6)$$

Take $i = 1$, $l = 2$, and $j = 2$. Then

$$\begin{aligned}\bar{P}_t(2, 2)\bar{U}_{0,t}(1, 2) &= \left[1 - \left(\frac{1}{4} - \frac{1}{4}e^{-4t}\right)\right] \int_0^t \left(1 - \left(\frac{1}{4} + \frac{1}{4}e^{-4u}\right)\right) du \\ &= \frac{1}{16} \left[9t + \frac{1}{2}e^{-4t} - \frac{3}{4} + 3te^{-4t} + \frac{1}{4}e^{-8t}\right]\end{aligned}$$

and

$$\begin{aligned}\bar{P}_t(1, 2)\bar{U}_{0,t}(2, 2) &= \left[1 - \left(\frac{1}{4} + \frac{1}{4}e^{-4t}\right)\right] \int_0^t \left(1 - \left(\frac{1}{4} - \frac{1}{4}e^{-4u}\right)\right) du \\ &= \frac{1}{16} \left[9t - e^{-4t} + \frac{3}{4} - 3te^{-4t} + \frac{1}{4}e^{-8t}\right].\end{aligned}$$

Now

$$16(\bar{P}_t(2, 2)\bar{U}_{0,t}(1, 2) - \bar{P}_t(1, 2)\bar{U}_{0,t}(2, 2)) = 6te^{-4t} - \frac{6}{4}(1 - e^{-4t}) = g(t) \quad (\text{say}).$$

Note that $g(0) = 0$ and $g'(t) = -24te^{-4t} < 0$, for all $t > 0$. This ensures that the condition (5.6) holds for $i = 1$, $l = 2$, and $j = 2$. For all the other choices of i , l , and j , the two sides of (5.6) are equal. Thus L_j possesses the IMIT property as well.

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